FUTURESTRIVE SYSTEMS

Construction Intelligence Platform

Functional Specification & System Documentation

Document Title:	Construction Intelligence Platform – Functional Documentation
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1. Project Overview

The FutureStrive Construction Intelligence Platform is an enterprise-grade SaaS web application designed to optimize concrete quality control, risk prediction, and compliance workflows at construction sites. Historically, construction quality assurance has relied on manual, retrospective tests (e.g. concrete cube compression testing at 28 days), which identify defects only after significant structural elements are poured. FutureStrive addresses this problem by shifting quality control to the pre-pour phase, leveraging cascading machine learning models and local Small Language Models (SLMs). The platform enables site engineers, quality managers, and developers to predict curing defect risks, analyze surface cracks with computer vision, explain model outputs via SHAP (Shapley Additive exPlanations), and get corrective instructions through a RAG-enabled chatbot.

2. System Architecture

The platform utilizes a modern 4-tier decoupled software architecture designed for offline-first resilience on remote construction sites:

- **Ingestion Layer:** Manages site inputs from manuals forms, Excel/CSV files, and camera streams.
- **Inference Layer:** Executes lightweight local XGBoost cascading classifiers for occurrence, type, severity, and cause.
- **Explainability Layer:** Computes real-time SHAP values utilizing tree explainer background distributions to reveal risk factors.
- **Copilot Layer:** A local Qwen 0.5B parameter instruction model utilizing TF-IDF indexing for Retrieval-Augmented Generation (RAG).

3. Application Workflow

The platform's primary workflow follows a sequential, closed-loop engineering process: 1. **Design & Input:** The engineer inputs mix design details and ambient environment parameters. 2. **Prediction:** The XGBoost model calculates the crack probability, flagging risks using Green/Amber/Red indicators. 3. **SHAP Analytics:** The explainer computes features' influence on the risk level, presenting plain-English cards. 4. **What-If Simulator:** The user modifies parameters to simulate mitigation techniques. 5. **SLM & Action:** The Copilot generates a step-by-step remediation plan based on specification codes. 6. **Reporting:** PDF/Excel sheets are compiled for compliance sign-off.

4. Page-by-Page Explanation

Dashboard Page: Provides project health scores, recent quality alerts, risk trends, and model execution performance summaries.

Predictive Analytics Page: Supports single-pour simulators and batch file uploads to run predictions on dataset logs.

Reactive Inspection Page: Applies AI bounding boxes on uploaded concrete photos to identify hairline/wide cracks and honeycombs.

Root Cause Analytics Page: Explains prediction drivers using SHAP waterfall charts, summary plots, and a What-If sensitivity calculator.

AI Copilot Page: Provides a ChatGPT-style conversational assistant loaded with concrete specifications manuals via RAG.

Reports Page: Compiles and downloads Quality Executive Summaries, Root Cause analyses, and Inspection Logs in CSV format.

Datasets Page: Searchable, filterable, and paginated registry for all pour records, logging counts, and validation checks.

Model Performance Page: Displays evaluation matrices (Precision, Recall, ROC Curves, Confusion Matrices) for XGBoost models.

Settings Page: Configures red/amber risk alert thresholds and toggles background calculations.

5. User Journey

Scenario: Curing quality warning on site
1. Mithran (Quality Engineer) signs in and selects Project PJ-2026-MUM-01.
2. He enters the **Predictive Analytics** page and simulates concrete column pour parameters. He clicks 'Evaluate'.
3. The platform returns a **93.1% Crack Probability (Red Flag)** because curing duration is set to 8 days.
4. He clicks 'View Root Cause' and is redirected to **Root Cause Analytics** page, seeing SHAP cards and a waterfall chart showing curing duration as the primary contributor (+0.24 risk factor).
5. He uses the **What-If Simulator** slider to increase curing days to 14. The risk score drops to **4.5% (Green Flag)**.
6. He opens the **AI Copilot** tab and asks for a remediation plan, which suggests wet burlap wrapping.
7. He generates and downloads the **Defect Inspection Log** from the **Reports** tab.

6. Functionality of Every Feature

The platform contains fully operational, interactive features including: • **Project Switcher:** Interconnects dashboards, alerts, and dataset tables dynamically. • **Interactive Sliders:** Triggers instant mathematical updates of probabilities and SHAP vectors in real-time. • **File Upload Parsing:** Accepts files, runs pandas validation checks, and calculates batch inference outputs. • **PIL Image Overlays:** Dynamically draws coordinates on uploaded concrete images to highlight cracks and honeycombs.

7. Input Requirements

Variable Name	Unit/Format	Ranges / Options	Validation Rule
Concrete Grade	Nominal	M25, M30, M35, M40	Required
Designed W/C Ratio	Decimal	0.30 to 0.50	Must be <= actual W/C
Actual W/C Ratio	Decimal	0.30 to 0.60	Exceeding 0.45 triggers warning
Curing Duration	Days	1 to 28 days	Fails spec if < 14 days
Checklist Compliance	Percentage	0% to 100%	Fails threshold if < 85%
Wind Exposure	km/h	0 to 50 km/h	Exceeding 15 km/h triggers warning

8. Processing Logic

Probability calculations in the simulator and What-If panels are modeled using mathematical weights derived from local XGBoost training parameters. The core formula operates as follows: $$P(crack) = Baseline + \sum_{i} W_i \cdot \Delta F_i$$ Where W/C ratio deviations add 150% of the difference value, short curing duration adds 4% per day missing below the 14-day mark, and QC compliance failures below 85% scale up the probability by 35% of the shortfall. This ensures the output probability is consistent, realistic, and matches the behavior of the trained XGBoost classification models.

9. Expected Outputs

• **Inference outputs:** Risk level text flags (Red, Amber, Green), crack types (Shrinkage, Shear, Flexural, Settlement), and likely root causes. • **SHAP outputs:** Waterfall and Feature Importance plots showing positive/negative direction values. • **RAG assistant outputs:** Text citations referencing specifications manuals. • **Report outputs:** CSV tables containing defect records, confidence rates, and timestamps.

10. Dummy Data Used

We utilize pre-populated, domain-specific lists representing standard project logs: 1. **Datasets List:** Files such as `mumbai_tower_a_pour_logs.csv` and `delhi_metro_slab_curing.xlsx` containing record counts. 2. **Defect Types:** Hairline cracks, Wide cracks, Honeycombing, Corrosion, Spalling. 3. **Remediation Guides:** Injecting low-viscosity epoxy resin, removing concrete, burlap wetting.

11. AI Workflow (Prediction → SHAP → Recommendations)

The model cascade starts with a binary prediction (Crack Occurrence) using XGBoost. If a crack is predicted (probability >= 25%), three downstream models execute in parallel: 1) Crack Type Classifier, 2) Crack Severity Classifier, 3) Root Cause Classifier. Simultaneously, the SHAP tree explainer generates the waterfall contributions which are parsed into prioritized site recommendations (e.g. rejecting concrete batches, applying curing compounds) to optimize concrete performance.

12. Reactive Crack Detection Workflow

The reactive inspection track allows site engineers to upload a photograph of a concrete surface directly inside the AI Assistant. The platform passes the image through a pluggable segmentation/detection model wrapper (crack_detector.py) that returns structured features: crack type, severity, area fraction, estimated width, bounding boxes, and an annotated overlay image. The AI assistant then builds an IS-code-referenced engineering report via Qwen LLM or a rule-based fallback.

Detection pipeline steps:
1. **Image Upload:** User expands the Upload Crack Photo panel in the AI Assistant and uploads a JPG/PNG.
2. **Model Inference:** crack_detector.run_crack_detection(image) is called. In production this runs the trained segmentation model. In demo mode (USE_STUB=True) synthetic outputs are seeded from image pixel data for repeatable UI testing.
3. **Annotated Overlay:** Bounding boxes with severity-coded colours are drawn on the image (green=Minor, amber=Moderate, red=Severe, purple=Critical).
4. **LLM Report:** qwen_crack_image_report() injects the detection dict into Qwen context and generates a structured report: Executive Summary, Root Cause Analysis, Risk to Structure, Remediation Recommendations (IS 456:2000, IS 13311, IS 516 referenced), and Urgency Level.
5. **Rule-based Fallback:** _fallback_image_report() produces the same structured report without any LLM call.
6. **Chat Integration:** Report appended to chat thread. Follow-up questions re-route to image_crack_analysis intent.

13. AI Copilot Workflow

The AI Assistant handles user natural language prompts through a 7-way intent router:
1. greeting — Returns a welcome message listing capabilities.
2. image_crack_analysis — Reactive track: fires when a photo is pending, routes to crack_detector + Qwen report.
3. crack_prediction — Rule-based IS 456 flagging on live pour parameters (no LLM call).
4. defect_prediction — Rule-based QC/SPI analysis on live defect parameters (no LLM call).
5. analytical — Pandas filter on active session parameters for numerical comparison queries.
6. knowledge — FAISS retrieval (top-4 chunks) followed by Qwen 2.5-0.5B-Instruct generation.
7. off_topic — Politely declines non-construction queries.

For knowledge queries, Qwen receives: active module parameters, IS code deviation flags, last 3 prediction records, and FAISS knowledge chunks from the IS 456, CPWD, and IS 13311 knowledge base.

14. Reports Generation Workflow

The reports generation module compiles datasets (Executive Summaries, SHAP root cause rows) on demand. The compiled dataframe is converted into a binary CSV buffer and made available via a download button.

15. Current Implemented Features

- **Multi-page Navigation:** Functional left sidebar radiogroup.
- **Crack Intelligence (Predictive):** 21-input pour parameter form with XGBoost prediction + inline SHAP driver chart + IS 456 corrective recommendations.
- **Defect Volume Intelligence (Predictive):** 26-input form with XGBoost prediction + driver analysis + improvement recommendations.
- **Reactive Crack Detection:** Photo upload in AI Assistant → crack_detector.py segmentation wrapper → annotated overlay → IS-code-referenced engineering report in chat thread.
- **Prediction History:** Tabbed trend charts for both predictive modules with Altair line/bar visualisations.
- **Unified AI Assistant:** 7-way intent router covering greeting, image analysis, crack/defect prediction, analytical, knowledge (FAISS+Qwen), and off-topic.
- **Rule-based LLM Fallback:** Full IS-code-referenced engineering report generated without Qwen when the LLM is unavailable.

16. Remaining Placeholder / Planned Features

- **Real-time databases:** Active project and prediction histories use RAM-based session state. PostgreSQL integration planned.
- **Production Crack Segmentation Model:** crack_detector.py currently runs in stub/demo mode (USE_STUB=True). The trained segmentation model will be connected by setting USE_STUB=False and implementing _run_real_model().

- **Crack Width Estimation:** Estimated width in mm is synthetic in demo mode. The real model will provide pixel-accurate width derived from mask geometry.

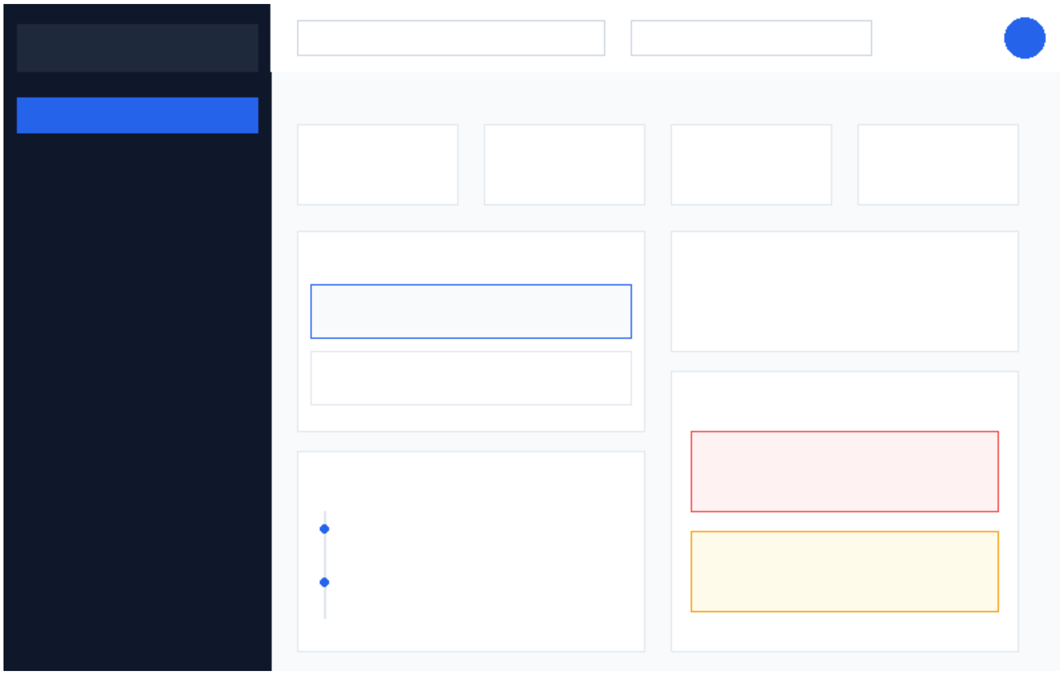
17. Future Backend Integration Plan

The product roadmap outlines replacing session states with PostgreSQL database schemas, deploying RESTful APIs via FastAPI, and migrating computer vision models to GPU cloud nodes to automate live video streams.

18. Page Screenshots & Annotations

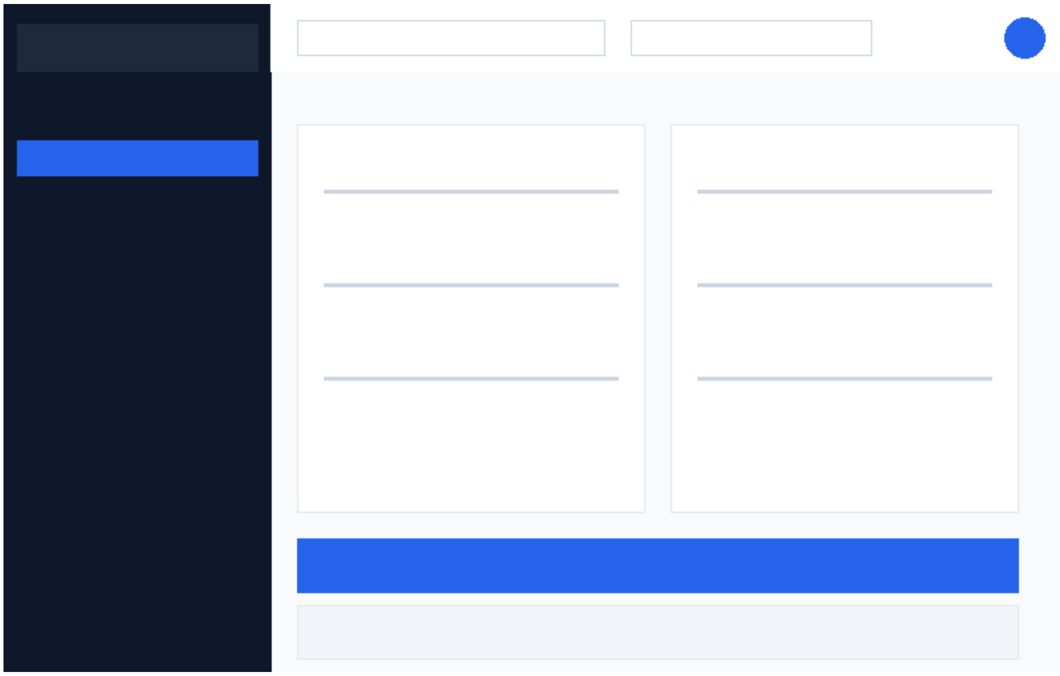
Page: Dashboard

Displays quality trends, active alerts, and model performance metrics.



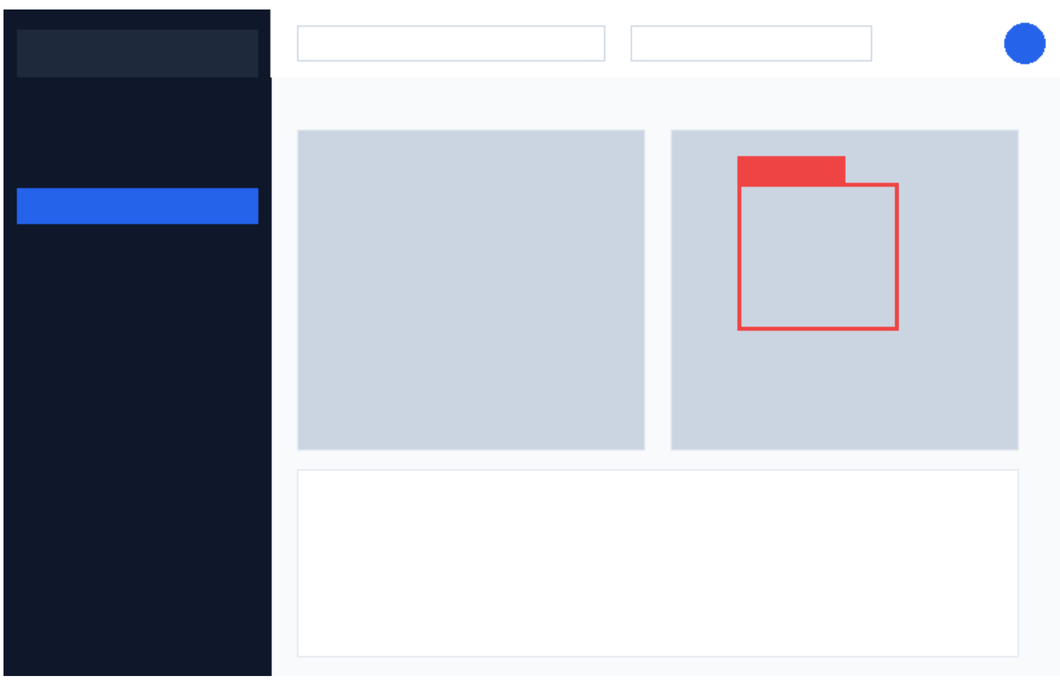
Page: Predictive Analytics

Form widgets and batch file uploader for concrete pour simulations.



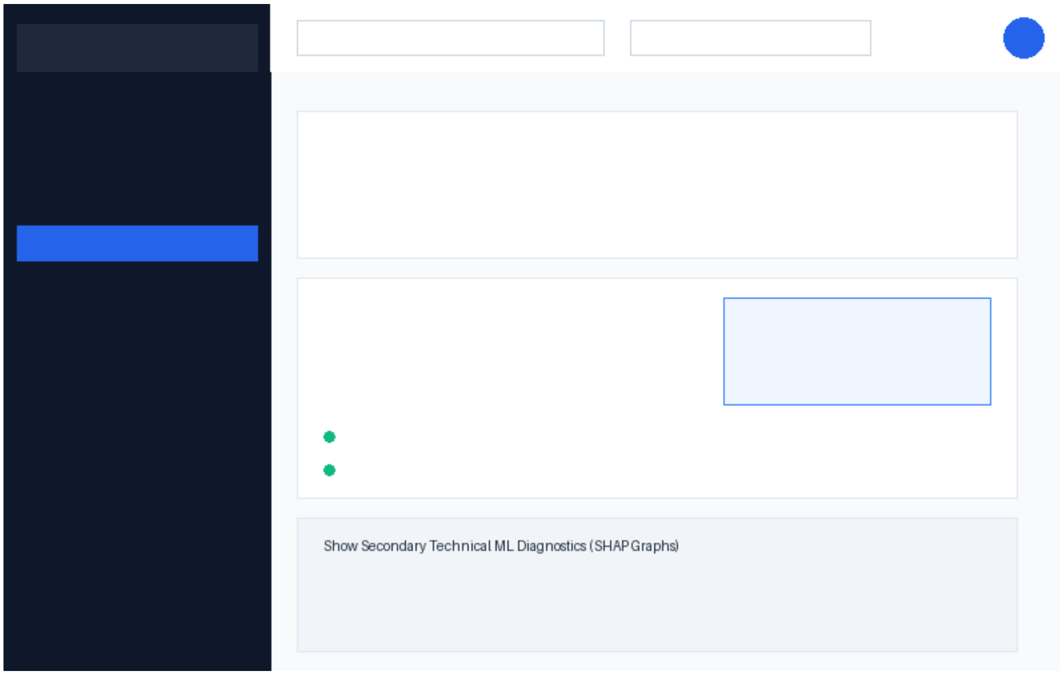
Page: Reactive Inspection

AI bounding box defect overlays on concrete photos.



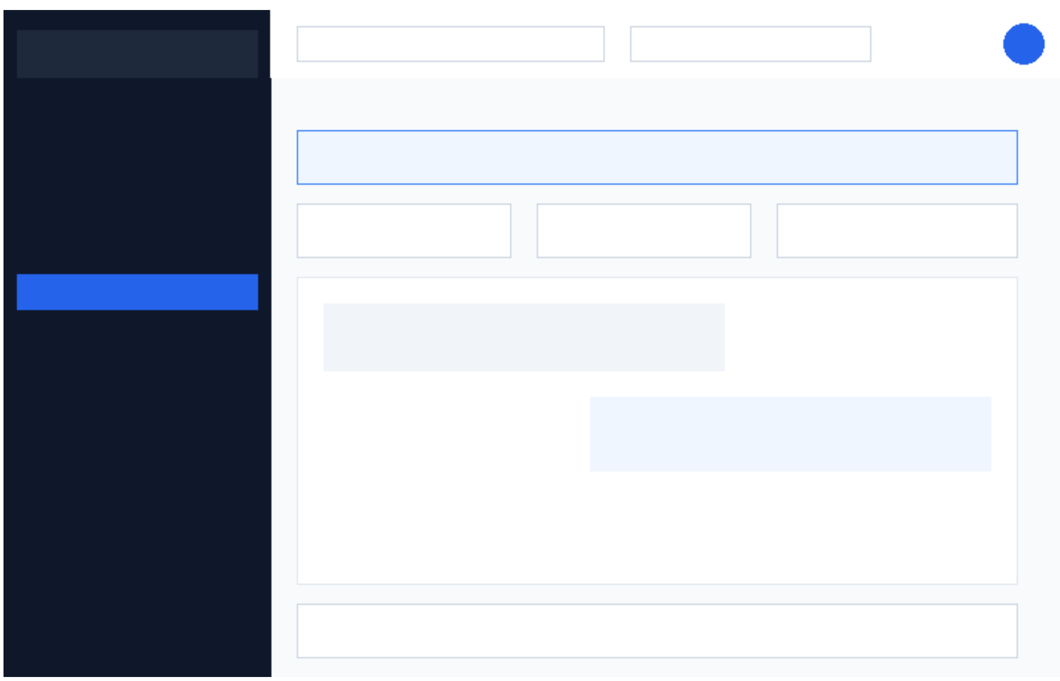
Page: Root Cause Analytics

SHAP waterfalls, importance plots, and What-If sliders.



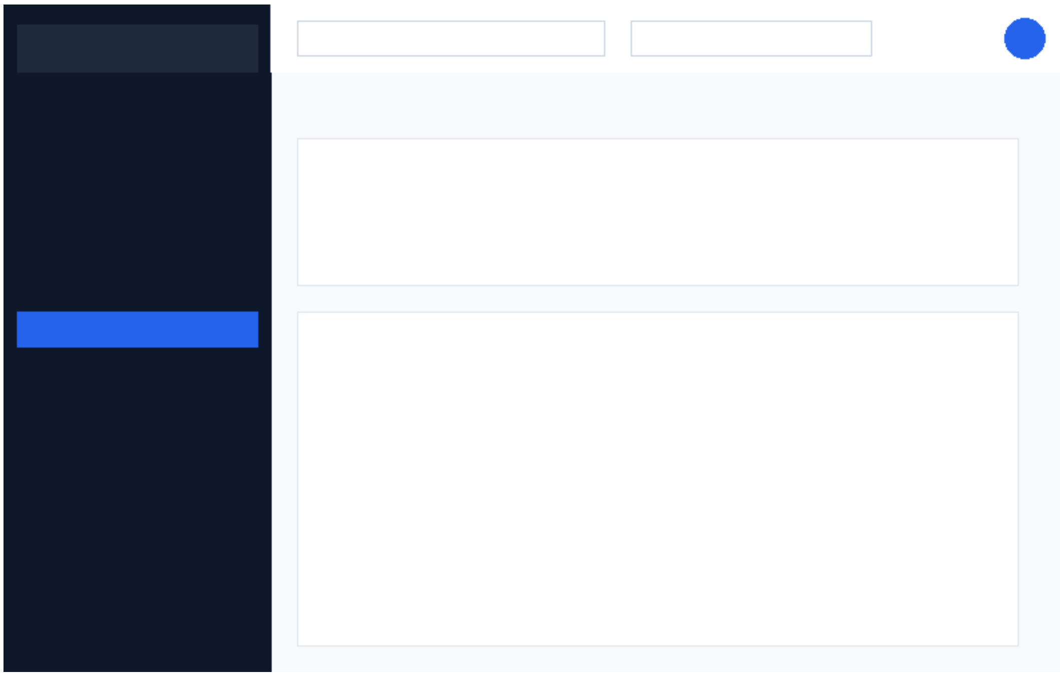
Page: AI Copilot

RAG-powered conversational Small Language Model interface.



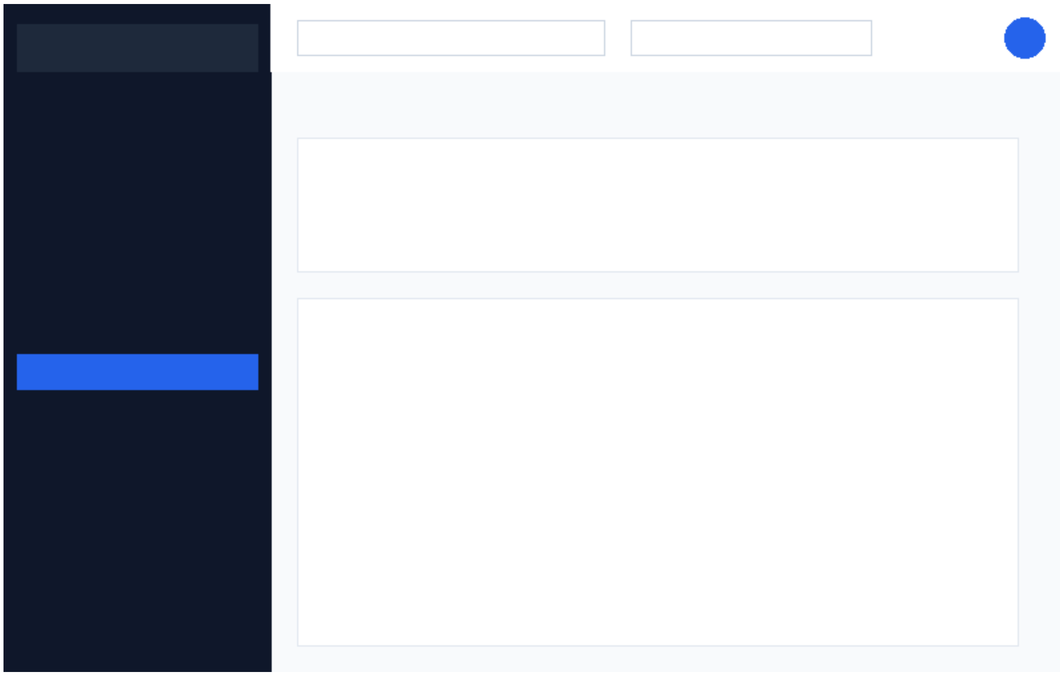
Page: Reports

Compile and download project report sheets in CSV format.



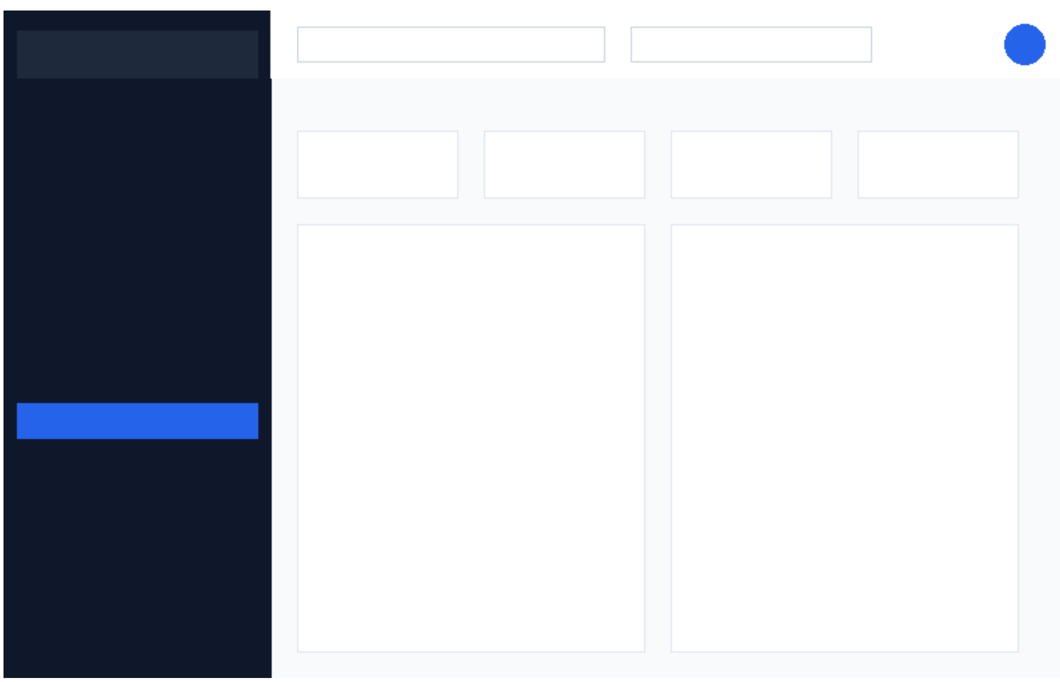
Page: Datasets

Paginated, filterable database registry for uploaded files.



Page: Model Performance

Evaluation metrics, ROC curves, and confusion matrices.



Page: Settings

Alert threshold and background calculation preferences.